MIL-H-85281A(AS)
15 July 1981
SUPERSEDING
MIL-H-85281(AS)
1 March 1979

MILITARY SPECIFICATION
HORIZON REFERENCE SET, SHIP MOUNTED, FOR LAMPS MK III HELICOPTER

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, design, and test requirements for the helicopter Horizon Reference Set (HRS).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-E-16400	Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore, General Specification
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment Including Ground Support Equipment
MIL-S-20708	Synchros, General Specification
MIL-L-24560/1	Light, Aircraft Warning, Red, 120 Volts, 90 Watts, Six Lamps, Symbol 160.1

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93) Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1710

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STANDARDS

MILITARY

MIL-STD-108	Definition of and Basic Requirement for Enclosures for Electric and Electronic Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-470	Maintainability Program Requirements for Systems and Equipments
MIL-STD-471	Maintainability Demonstration
MIL-STD-710	Synchros, 60 and 400 Hertz
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-1378	Requirements for Employing Standard Electronic Modules
MIL-STD-1399/ 103	Interface Standard for Shipboard Systems Section 103 Electric Power, Alternating Current
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-2074	Failure Classfication for Reliability Testing
MIL-STD-45662	Calibration Systems Requirements

PUBLICATIONS

NAVAL AIR SYSTEMS COMMAND

AS-4613	Application and Derating Requirements for Electronic Components

DRAWINGS

NAVAIRENGCEN DRAWING

619370	Horizon Reference Set A/W 37A-1

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

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3. REQUIREMENTS

3.1 System Description. The HRS, when installed in accordance with this specification, shall provide a horizon reference which is independent of ship's rolling motion. The visual display shall be a roll-stabilized lighted reference bar to be located in the vicinity of the top of the hangar face of LAMPS MK III ships such as to be in the same plane of vision as seen by a LAMPS MK III SH-60B helicopter pilot conducting a normal obstruction clearance scan during landing approach and recovery operation. The set shall be composed of three separate units:

a. A Stabilization Data Indicator Assembly composed of the lighted reference bar and the bar drive mechanism.

b. An Electronic Component Assembly containing the control electronics, operating controls and indicators.

c. A Control-Indicator containing duplicate operating controls and indicators of the Electronic Component Assembly for remote control of the set.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.2.3 and 6.4).

3.3. Characteristics. HRS equipment shall, except as modified herein, meet all characteristics specified in the general equipment specification, MIL-E-16400 and the requirements specified herein.

3.3.1 Performance Characteristics. The HRS, when combined into an operating configuration, shall conform to the following performance requirements.

3.3.1.1 Stabilization Accuracy. The horizon reference bar shall be stabilized through a roll amplitude up to the set normal operating limits (see 3.2.1.3) to an accuracy within ±0.5° of the input signal.

3.3.1.2 Servo drive characteristics. The maximum angular velocity of the servo system driving the reference bar shall not be less than 20° per second and the maximum angular acceleration under the same conditions shall not be less than 15° per second per second. The servo system damping ration shall not be less than 0.70 and not be more than 0.95. The servo system shall have sufficient torque to withstand the wind velocities specified in MIL-E-16400.

3.3.1.3 Roll amplitude limits. The reference bar electro-mechanical drive shall incorporate roll amplitude limiting provisions which shall be adjustable. These adjustable limits shall provide for:

a. A minimum normal operating limit of not more than ±17° and a maximum normal operating limit of not less than ±30° from the installation vertical. Adjustment increments between the minimum and maximum limits shall not be greater than 1°.

b. Inhibiting circuitry to limit reference bar rotation more than 1° greater than the set normal operating limit.

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c. Positive reversal drive force should the reference bar rotation be more than 3°±1° greater than the set normal operating limit.

d. Mechanical stops to prevent further reference bar rotation beyond more than 8°±1° greater than the set normal operating limit.

3.3.1.4 System Secure. The electro-mechanical drive system shall automatically align and lock the reference bar in a position parallel to the ship's deck when the HRS is shut down.

3.3.1.5 Reference Bar Lighting. The reference bar shall be illuminated using electroluminescent lamps conforming to 3.4.3. The length of electroluminescent lighting on the bar shall not be less than 9 feet 6 inches and the width shall not be less than 2 inches; lighting discontinuities between adjacent lamps shall not be greater than 1 inch. The intensity of the reference bar lighting shall be smoothly controllable from zero to maximum.

3.3.1.6 Fault detection and warning. The HRS shall incorporate provisions to monitor potential faults and provide automatic shut down and warning as follows.

3.3.1.6.1 Fault detection. Monitoring shall detect the existence of any of the following faults:

a. Partial or complete loss of the input signal from the ship's gyro.

b. Partial or complete loss of the signal from the amplifier to the servo motor.

c. A differential of greater than two angular degrees for more than four seconds between the input signal and the reference bar position when the reference bar is within the range of normal operating limits (see 3.3.1.3).

d. Loss of input power to the HRS control circuitry.

e. Partial or complete failure of the internal power supply.

3.3.1.6.2 Automatic shutdown. The presence of any of the system faults listed in 3.3.1.6.1 shall cause the HRS to automatically perform the following sequence:

a. Immediately turn off the reference bar illumination,

b. Immediately turn on the red warning lamp,

c. Align and lock the reference bar in position parallel to the installation platform if the servo system is operating, and

d. Remove power from the servo system after 10 seconds.

3.3.1.6.3 Warning lamp. The warning lamp shall consist of an assembly conforming to MIL-L-24560/1 (formerly BuShips Drawing 9000-S6405-74231, Revision F) which shall have its primary location atop the housing of the Stabilization Data Indicator Assembly. The cable to the warning lamp assembly

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shall be a minimum of 2 feet long to allow for alternate installation of the warning lamp assembly on the Stabilization Data Indicator Assembly foundation (see 3.6.4). The intensity of the warning lamp shall be controllable from full brilliance down to no less than 25% of the foot lambert output at full brilliance.

3.3.1.7 Operating control and indicators. The Electronic Components Assembly and the Control-Indicator shall have the following switches, controls and indicators:

a. Power switch with local/remote indicators

b. Start switch with indicator

c. Stop switch with indicator

d. Reference bar stabilized indicator

e. Reference bar locked indicator

f. Fault indicator and reset switch

g. Reference bar illumination intensity control

h. Warning lamp intensity control

i. Panel and indicator illumination intensity control

In addition the Electronic Component Assembly shall have the following:

j. Dial display of reference bar position

k. Test controls and indicators

l. Indicators for 60Hz and 400 hz power and for gyro signal reference voltage.

3.3.1.7.1 Local/remote control. Design of the HRS shall provide for control of the installed set from either the Electronic Component Assembly or the Control-Indicator if installed. Control system design shall not require the installation of Control-Indicator to achieve system operation and specified performance.

3.3.2 Physical characteristics.

3.3.2.1 General specification. The HRS shall meet all of the physical characteristics specified in the general specification MIL-E-16400 and the characteristics specified herein.

3.3.2.2 Physical size and weight. The HRS equipment shall physically conform to the installation and interface criteria of NAVAIRENGCEN Drawing 619370.

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3.3.2.3 Modular construction. The Electronic Components Assembly shall employ modular construction techniques conforming to the Standard Electronic Module requirements of MIL-STD-1378. The selection of specific assembly design and packaging techniques shall reflect the reliability and maintainability requirements specified herein.

3.3.3 Reliability.

3.3.3.1 Reliability Program. The reliability program shall be conducted in accordance with MIL-STD-785. A reliability demonstration plan shall be developed to verify conformance with 3.3.3.2 (see 4.7.8).

3.3.3.2 Mean Time Between Failure (MTBF). MTBF, defined as the total operating time in hours, divided by the total number of failures, shall equal or exceed 500 hours.

3.3.3.3 Failure investigation, analysis and corrective action system. A closed loop failure data collection, analysis and corrective action systems, in accordance with MIL-STD-785 shall be developed and employed for all failures (see 4.6.2).

3.3.3.4 Failure reports. All relevant and non-relevant failures shall be reported. A failure report containing the details of failure investigation shall be prepared (see 4.6.2.4) for each failure occurring during contractually required testing.

3.3.4 Maintainability.

3.3.4.1 Maintainability Program. The maintainability program shall be conducted in accordance with MIL-STD-470. A maintainability demonstration plan shall be developed to verify conformance with 3.3.4.2 and 3.3.4.3 (see 4.7.9.).

3.3.4.2 Mean time to repair (MTTR). For organizational maintenance, the MTTR shall not exceed 15 minutes. Maximum maintenance time for correction of a failure shall not exceed 45 minutes at the 95th percentile of distribution. MTTR is defined as the total corrective maintenance time divided by the total number of corrective maintenance actions during a given period of time.

3.3.4.3 Maintenance support. No scheduled preventive maintenance actions other than periodic inspection, cleaning, lubrication, corrosion prevention or replacement of electroluminescent lamps shall be required. No corrective or preventive maintenance action shall require more than two persons for its accomplishment.

3.3.4.4 Maintenance procedures. A complete maintenance procedure shall be developed using the support equipment and tools approved for organizational maintenance of the HRS. The maintenance procedures shall be included in the technical manual.

3.3.5 Environmental Service Condition. HRS equipment shall conform to the environmental service condition requirements of MIL-E-16400. Interpretation those requirements shall be in accordance with the following.

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3.3.5.1 Exposed equipment. The Stabilization Data Indicator, including the illuminated reference bar and the warning lamp, shall be classified as exposed equipment.

3.3.5.2 Sheltered equipment. The Electronic Components Assembly and the Control-Indicator shall be classified as sheltered equipment.

3.3.5.3 Temperature ranges. Temperature range 2 shall apply for exposed equipment and temperature range 4 shall apply to sheltered equipment.

3.3.5.4 Gunblast. Not applicable.

3.3.5.5 Sunshine. Delete MIL-E-16400 requirement. The electroluminescent lamps shall maintain specified performance and show no evidence of physical deterioration after the sunshine test of 4.7.3.6.

3.4 Parts, materials and processes. Except as specified herein, the selections and use of parts, material and processes for HRS equipment shall be in accordance with the general specification MIL-E-16400.

3.4.1. Derating. Derating of electronics and electrical parts and materials shall be accomplished in accordance with AS-4613, Class A.

3.4.2 Use of Cadmium. Electroplating or mechanical deposition of cadmium shall not be used.

3.4.3 Electroluminescent lamps. Electroluminescent lamps used on the reference bar shall consist of a thin luminious element incased in a flat enclosure of clear acrylic or fiberglass with sealed leads for connection to the electrical supply.

3.4.3.1 Electrical requirements. The lamps shall satisfactorily operate from a supply of 60Hz to 400 Hz with voltage variable up to 130 volts maximum.

3.4.3.2 Color. Each lamp shall emit green light with the peak emissions at a wavelength of 540 nanometers ±3% when operated at an ambient temperature of 23° ±10°C (73° ± 18°F) from a 120V 400 Hz supply. There shall be no significant color change with changes in supply voltage.

3.4.3.3 Luminance. Lamp luminance, when operated at an ambient temperature of 23° ±10°C (73° ± 18°F) from a 120V 400Hz supply, and after a five-hour stabilization period, shall have an initial luminance of not less than 18 foot lamberts, a luminance of not less than 8 foot lamberts after operating 1000 hours, and a luminance of not less than 4 foot lamberts after operating 3000 hours. The luminance at any point of the luminious area of the lamp shall not exceed the luminance at any other point by more than 15%.

3.4.3.4 Lamp Construction. Each lamp shall consist of the electroluminescent element housed in a flat enclosure of clear acrylic or fiberglass capable of being bolted to a flat surface without undue strain on either the lamp or the mounting bolts. Lamp electrical leads shall be shrouded in sealant, projecting not more than 5/8", at the point of entry into the lamp. Lamps and electrical lead terminations shall be hermatically sealed as defined in MIL-STD-108.

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3.5 Electrical design and construction. Equipment of the HRS shall conform to the electrical design and constructions requirements of MIL-E-16400 with the following interpretations.

3.5.1. Electrical power. The HRS shall be designed to operate from ac electrical power having characteristics as specified in section 103 of MIL-STD-1399 for 400 Hz type I and for 60Hz type I.

3.5.2 Vertical reference signal. The HRS shall be capable of accepting its roll input signal from a type 18CX4 400Hz synchro control transmitter as specified in MIL-STD-710 and MIL-S-20708 whose output is two electrical degrees for each degree of actual ship roll.

3.6 Mechanical design and construction. Equipment of the HRS shall conform to the mechanical design and construction requirements of MIL-E-16400 with the following interpretations.

3.6.1 Degree of enclosure. The degree of enclosure for exposed equipment (see 3.3.5.1) shall be watertight and for sheltered equipment (see 3.3.5.2) shall be dripproof (45 degrees).

3.6.2 Hazardous atmosphere. Protection from hazardous atmosphere is not required.

3.6.3 External connections. The method and placement of external connections for HRS equipment shall be in accordance with NAVAIRENGCEN Drawing 619370.

3.6.4 Equipment mounting. Equipment mounting provisions shall be in accordance with NAVAIRENGCEN Drawing 619370. The foundation for the Stabilization Data Indicator Assembly shall not be supplied as part of the HRS set unless specified in the contract. The Electronic Components Assembly and the Control-Indicator shall be designed for bulkhead mounting.

3.6.5. Control, indicators and panel layouts. Controls and indicators for the Electronic Components Assembly and for the Control-Indicator shall functionally conform to 3.3.1.7; panel illumination is required. In addition, the panel of the Control-Indicator shall be designed for dark adapted areas. Panel layouts shall be in general conformance with the referenced outline drawings of NAVAIRENGCEN Drawing 619370. The Electronic Components Assembly shall include a time totalizing meter to indicate elapsed time for set operation and a key switch to disable the servo drive of the Stabilization Data Indicator Assembly for maintenance safety.

3.7 Thermal design. Thermal design shall be in accordance with the requirements of MIL-E-16400. Equipment design shall not require forced air or water cooling.

3.8 Identification and marking. Nomenclature, serialization and other equipment identification shall be in accordance with MIL-N-18307. Functional labeling on equipment shall conform to MIL-STD-1472. Conductor identification and terminal end identification internal to equipment shall conform to MIL-E-16400.

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3.9 System safety program. A system safety program shall be developed and maintained in accordance with MIL-E-16400 for all design efforts as well as production and testing of the equipment. The safety criteria to be applied during equipment hardware design, selection and construction shall include the criteria specified in MIL-E-16400.

3.10 Workmanship. Workmanship shall conform to requirements 5 and 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. Unless otherwise specified in this section the quality assurance provisions shall be in accordance with the general specification MIL-E-16400.

4.2 Classification of inspection. The methods of inspection fall within the following classifications:

a. Initial performance tests

b. Operational burn-in

c. First Article Inspection

d. Quality conformance inspection

(1) Production inspection
(2) Production control inspection

4.2.1 Initial performance test. After completion of manufacturing, assembly and production adjustment operations, and prior to proceeding with any of the other inspections of 4.2, each item shall be operated under the standard ambient conditions (see 4.5) and a record made of all data necessary to determine compliance with required system performance (see 3.3.1). These data shall provide the criteria for checking satisfactory performance of the item either during or at the conclusion of operational burn-in.

4.2.2 Operational burn-in. After completion of initial performance tests each item shall be operated for a period of time or a number of cyclic operations, under stressing conditions sufficient to expose infant mortality type failures. Operational burn-in shall be completed on each item before commencement of first article inspection or quality conformance inspection. The duration employed shall be based on the susceptibility to infant mortality of the technology employed in the item. The contractor shall include the proposed burn-in duration and stressing conditions in the test plan prepared in accordance with 4.3. The duration of the burn-in for each item shall be subject to further adjustment during production, with procurement agency approval, to obtain optimum effectiveness commensurate with cost.

4.2.3 First article inspection. - The item(s) designated for first article inspection as specified (see 6.2), shall be subjected to examinations and tests necessary to determine compliance with this specification. Unless otherwise specified in the contract (see 6.2), first article inspection shall include examinations and tests shown in table I and shall be performed as approved in accordance with 4.3.

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4.2.4 Quality conformance inspection.

4.2.4.1 Production inspection. A production inspection shall be made on each item offered for delivery. This inspection shall include such examination and tests as will prove the workmanship and reveal omissions and errors of the production process such as, functional and performance test at a limited number of points in the required range, tests which detect deviations from design, tests of controls/adjustments, and tests which detect hidden defects of materials. Unless otherwise specified in the contract (see 6.2), production inspection shall include the examination and tests shown in Table I and shall be performed as approved in accordance with 4.3.

4.2.4.2 Production control inspection. Production control inspection shall be conducted on a sampling basis when specified in the contract or order (see 6.2). This inspection shall consist of the examination and tests which encompass functional and performance tests throughout the entire range of operation; tests which will detect any deterioration of the design by wear of such items as dies, molds, and jigs, and by substitution of different parts; tests which detect deviations in the processing of the materials; tests to determine temperature rise produced in operation and ability of equipment to withstand this heat; tests of efficiency; and tests of the performance with other equipment in a system. These tests shall be performed on the complete equipment as offered for delivery. When ordered (see 6.2), production control inspection shall include the examination and tests shown in Table I and shall be performed as approved in accordance with 4.3.

4.3 Examination and test plans. The contractor shall submit written examination and test plans which contain the procedures, data forms, a list of test equipment to be used and the sequence of test for performing operational burn-in, first article and quality conformance inspections to the procuring activity via the Government inspector for approval prior to beginning inspection as specified in the CDRL (see 6.3). Examination and tests shall be identified in accordance with 4.2.

4.3.1 Test procedures. Test procedures shall contain step-by-step information for each inspection, test and demonstration to be performed. The test procedures shall include preparation for test, connections, adjustments, measurements, including test parameters and tolerance limits, and disconnection of item under test.

4.3.1.1 Test equipment listing. The test equipment list shall identify the test equipment to be used by name, model, range and accuracy. Special test equipment shall also be listed and identified as to its capability to correctly perform the tests.

4.3.1.2 Test data forms. Test data forms shall be prepared for each of the procedures. These forms shall be organized to indicate the acceptance test limits of the test and the actual readings observed. In addition, the form shall indicate a statement of acceptance or rejection after each test. When test data forms are specified by the referenced specification, they shall be utilized. The form shall contain a signature and date block for certification of the data by the test engineer.

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TABLE I Examination and tests

Subject	Requirement	Test Method	First Article	Quality Conformance Inspection	
				Production Inspection	Production Control Inspection
Examination					
Weight	3.3.2.2	4.7.1	X	X	
Size	3.3.2.2	4.7.1	X	X	
Parts & Materials	3.4	4.7.1	X	X	
Finish	3.4	4.7.1	X	X	
Marking	3.8	4.7.1	X	X	
Safety	3.9	4.7.1	X	X	
Workmanship	3.10	4.7.1	X	X	
Initial Performance		4.2.1	1/	1/	
Burn-In		4.2.2	1/	1/	
Performance					
Full System	3.3.1	4.7.2.1	X	X	
Individual Item		4.7.2.2		X	
Environmental					
Temperature	3.3.5	4.7.3.1, 4.7.3.2 & 4.7.3.3	X		X
Humidity	3.3.5	4.7.3.4	X		
Salt Fog	3.3.5	4.7.3.5	X		
Sunshine	3.3.5	4.7.3.6	X		
Fungus	3.3.5	4.7.3.7	X		
Wind	3.3.5	4.7.3.8	X		
Icing	3.3.5	4.7.3.9	X		
Shock	3.3.5	4.7.3.13	X		
Vibration	3.3.5	4.7.3.14	X		X
Dust (fine sand)	3.3.5	4.7.3.17	X		
Enclosure	3.6.1	4.7.4	X		X
Electrical	3.5	4.7.5	X		X
Reliability Demonstration	3.3.3	4.7.8	X		X
Maintainability Demonstration	3.3.4	4.7.9	X		

1/ Initial performance tests and operational burn-in shall be completed prior to start of first article examination or production inspection.

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4.4 Test reports.

4.4.1 Initial performance and operational burn-in reports. Test data sheets for initial performance tests and for operational burn-in tests shall be included in the test reports of 4.4.2 or 4.4.3 as applicable for the specific units tested.

4.4.2 First article inspection report. A first article inspection report containing results of all tests shall be prepared as specified in the CDRL (see 6.3). When the tests are performed in accordance with referenced documents, the test data and test data forms shall comply with these referenced documents.

4.4.3 Quality conformance inspection reports. Quality conformance inspection reports containing the results of all examinations and tests performed shall be prepared as specified in the CDRL (see 6.3). The report shall list the serial numbers of the equipment tested, and the date tested. When tests are performed in accordance with referenced documents, the test data and test data forms shall comply with these referenced documents.

4.5 Test facilities, chambers, apparatus and standard test conditions. Test facilities, chambers, apparatus and standard test conditions used in conducting the tests specified herein shall be in accordance with the general specification MIL-E-16400, except that paragraph 4.5.2 (a) of MIL-E-16400G is modified to read:

(a) Conform to laboratory standards whose calibration is traceable to a calibration source in accordance with MIL-STD-45662.

4.6 Performance of test. Performance of tests shall be in accordance with MIL-E-16400 and the following:

4.6.1 Pre-performance test. The initial performance test of 4.2.1 shall constitute the pre-performance test relative to operational burn-in (see 4.2.2).

4.6.2 Failures. Unless otherwise specified in the individual test requirements, failures which occur during required testing shall be identified, classified, investigated, analyzed, corrected and recorded in accordance with the following:

4.6.2.1 Failure criteria. Equipment, or portions thereof, subjected to a test specified herein shall be considered to have failed when any of the following occur:

a. Performance parameters exceed limits specified.

b. Catastrophic or structural failure.

c. Distortion or displacement of mechanical parts that causes difficulty of servicing or replacing a part.

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d. Any condition that results in a hazard to personnel safety.

e. Deterioration, corrosion, or change in performance limits causing failure to meet operational service or maintenance requirements.

f. Deterioration of sealing compounds that would cause a decrease in service life or reliability.

g. Any condition that requires or results in a corrective maintenance action.

4.6.2.2 Failure classification. All failures shall be classified as relevant or nonrelevant in accordance with MIL-STD-2074.

4.6.2.3 Failure investigation, analysis, and corrective action. A closed loop failure data collection, analysis, and corrective action system, in accordance with MIL-STD-785 must be employed for all failures.

4.6.2.4 Failure reporting. A report of failures occuring during or as a result of tests specified herein shall be included in the reports of 4.4.2 and 4.4.3, together with the probable cause of failure and corrective action taken.

4.7 Test methods. The following test methods of MIL-E-16400 are applicable, not applicable or modified as specified.

4.7.1 Surface examination. Applicable.

4.7.2 Performance test. The equipment shall be subjected to a performance test including all operating controls to determine compliance with the performance parameters and safety requirements herein.

4.7.2.1 Full system performance test. HRS equipment, installed in accordance with NAVAIRENGCEN Drawing 619370 and combined into a system configuration, shall be subjected to a performance test to determine compliance with 3.3.1; tests shall be performed as approved in accordance with 4.3.

4.7.2.2 Individual item performance test. Individual equipment items not delivered as part of a complete equipment set shall be subjected to a functional and performance test at a limited number of points in the required operating range to detect deviations from design. Operating test points shall include the full range of operating controls/adjustments installed in the item. The test shall be performed as approved in accordance with 4.3.

4.7.3 Environmental tests.

4.7.3.1 Temperature test sequence. Applicable.

4.7.3.2 Low temperature. Applicable.

4.7.3.3 High temperature. Applicable.

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4.7.3.4 Humidity. Applicable.

4.7.3.5 Salt fog. Applicable. Exposed equipment shall be tested in accordance with Method 509.1, Procedure I of MIL-STD-810. Electrical operation is required at the end of the exposure period. If available test chambers preclude testing with the full size reference bar testing may be conducted with a shorter bar representative of production fabrication with procuring activity approval.

4.7.3.6 Sunshine. Tests of complete assemblies not required. The electroluminescent lamp employed on the reference bar shall be subjected to the sunshine test of Method 505.1, Procedure I of MIL-STD-810. In addition to the General Requirements of MIL-STD-810, the lamp shall show no post test evidence of deformation, surface instability, splits, crazing, increased brittleness or break in lamp seal.

4.7.3.7 Fungus. Applicable.

4.7.3.8 Wind velocity. The Stabilization Data Indicator Assembly, with reference bar and warning lamp installed, shall be suitably mounted for operation with impinging wind forces. The equipment shall be operated with a sinusoidal input signal providing roll amplitudes of ±30 degrees at a frequency of 5 cycles per minute while being subjected to a wind velocity of 75 knots directed normal to the reference bar in the horizontal plane. The equipment shall operate normally and not incur damage during continuous operations under these conditions for one hour. Upon successful completion of this test and the post-test inspection, the Stabilization Data Indicator Assembly, non-operating, shall be subjected to a wind velocity of 100 knots directed normal to the reference bar in the horizontal plane for a period of 30 minutes. Upon completion the equipment shall be visually inspected for damage and then subjected to a performance test.

4.7.3.9 Icing. The Stabilization Data Indicator Assembly with reference bar and warning lamp installed, in a non-operating condition, shall be subjected to a fresh water ice forming spray until glaze ice build-up has reached 4.5 pounds per square foot of exposed surface. The ice build-up shall be maintained for a period of four hours. The temperature will then be increased to permit normal melting. Upon completion the equipment shall be visually inspected and subjected to a performance test.

4.7.3.10 Hydrostatic pressure. Not applicable.

4.7.3.11 Underwater explosion. Not applicable.

4.7.3.12 Gun blast and nuclear air blast. Tests not applicable.

4.7.3.13 Shock. Applicable. The Stabilization Data Indicator Assembly, with reference bar and warning lamp installed, shall be mounted on a pedestal (NAVAIRENGCEN part number 621154-1 or equivalent).

4.7.3.14 Vibration. Applicable. The Stabilization Data Indicator Assembly, with reference bar and warning lamp installed, shall be mounted on the same pedestal as used for the shock tests (see 4.7.3.13).

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4.7.3.15 Inclination. Not applicable.

4.7.3.16 Magnetic field test. Not applicable.

4.7.3.17 Dust (fine sand). The Stabilization Data Indicator Assembly, with reference bar and warning lamp installed shall be subjected to the dust tests as specified in MIL-E-16400; no change in orientation to the equipment shall be required during the test. If available test chambers preclude testing with full size reference bar testing may be conducted with a shorter bar representative of production fabrication with procuring activity approval.

4.7.3.18 Accelerated life tests. Not applicable.

4.7.3.19 Paint system test. Not applicable.

4.7.4 Mechanical and structural tests. Only the enclosure tests of MIL-E-16400 are required.

4.7.5 Electrical tests. The electrical tests shall be in accordance with MIL-E-16400.

4.7.6 Electromagnetic interference characteristics. Tests not required.

4.7.7 Thermal test. Tests not required.

4.7.8 Reliability demonstration. A reliability demonstration shall be conducted in accordance with MIL-STD-781 and as specified herein. The demonstration shall be conducted subsequent to full system performance tests and in accordance with the demonstration test plan approved in accordance with 4.3.

4.7.8.1 Demonstration criteria. Satisfactory demonstration of the reliability requirements (see 3.3.3) is contingent on achievement of the following test time and maximum failures.

Test time, hours	Failures not to exceed
1945	1
2650	2
3350	3

Demonstration test shall be conducted under existing outdoor environmental conditions with the Stabilization Data Indicator Assembly exposed to weather. The other components of the system shall be sheltered to an extent simulating the shipboard installation with minimum shielding from outdoor temperature excursions. The test shall be conducted at nominal input voltages and frequencies and without vibration stressing. The equipment will be in a power-off condition for a minimum of two hours continuously during each calendar day of testing and shall be cycled on and off the equivalent of once per three hours of operation. During periods of equipment operation, control time shall be divided between the Electronic Components Assembly and Control-Indicator in a manner to be described in the test plan/procedure. Inputs to the system shall simulate expected ship roll motion amplitudes and frequency.

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4.7.8.2 Performance check. A complete performance test, as approved in the demonstration test plan, shall be performed on each system under test at the start of the demonstration and at the conclusion of the demonstration. During the demonstration a daily check shall be performed on each set in the demonstration; the daily check shall consist of an abbreviated performance test using selected performance parameters as approved in the demonstration test plan. Each daily check shall include tests for proper functioning of the fault detection and warning features and the roll limiting features of the set.

4.7.8.3 Maintenance. Preventive maintenance during the reliability demonstration shall be limited to that planned and approved for operational usage, and shall be identified in the demonstration test plan. Suspicion of incipient failure shall not be justification for replacing a part prior to its actual failure. If the failure of a suspected part may cause catastrophic failure, the part may be replaced with the approval of the procuring activity; a part so replaced shall be treated as a failed item and the catastrophic failure shall be assumed to have occurred. All preventive and corrective maintenance actions shall be recorded as demonstration data.

4.7.9 Maintainability demonstration. The contractor shall conduct a formal maintainability demonstration to assure the maintainability requirements of this specification (see 3.3.4). The demonstration shall be performed in accordance with MIL-STD-471A, Test Method 8 on one of the first three production sets, utilizing maintenance procedures, support equipment and tools approved for use with the system. Corrective maintenance that occurs as a result of failures during other First Article Testing may be used toward this demonstration requirement if performed in accordance with approved maintenance procedures and properly documented.

4.8 Disposition of tested equipment. Unless otherwise provided for in the contract, equipment which has undergone first article inspection (see 4.2.3 and 6.2) or production control inspection (see 4.2.4.2) shall be refurbished as necessary to return to satisfactory operating condition. Refurbishment shall include the replacement of all failed parts; parts which have deteriorated but do not exceed specified tolerances shall also be replaced unless the procuring activity directs otherwise. The equipment shall successfully complete the applicable acceptance tests prior to shipment. Unless otherwise specified in the contract, the contractor shall furnish all replacement parts required for refurbishment and shall maintain a record of restoration actions. Disposition of equipment deemed non-repairable shall be as approved by the procuring activity.

5. PACKAGING

5.1 Horizon Reference Sets. Unless otherwise specified (see 6.2), preservation, packaging, packing and marking for shipment of all procured complete Horizon Reference Sets shall be performed in accordance with MIL-STD-794, Level "C".

5.2 Spares.

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5.2.1 Preparation. All designated spare assemblies or subassemblies shall be functionally tested, "burned-in" and adjusted where required to allow direct item replacement. Items that require in-system fine tuning shall be conspicuously marked with an appropriate instructional placard.

5.2.2 Packaging and marking. Unless otherwise specified (see 6.2), preservation, packaging, packing and marking for shipment of all procured spares shall be performed in accordance with MIL-STD-794, Level "A".

6. NOTES

6.1 Intended Use. The Horizon Reference Set to be supplied in accordance with this specification is intended to provide the capability to operate LAMPS MK III helicopters in adverse weather conditions from the small flight decks of LAMPS MK III ships experiencing extreme motions.

6.1.1 General LAMPS mission statement. A requirement exists to provide the capability to operate helicopters in all weather conditions, up to and including upper Sea State 5, from the decks of nonaviation ships operating within the LAMPS mission profile.

6.1.2 Mission profile. LAMPS MK III helicopters will be recovered aboard LAMPS ships first by approaching from astern, flying to arrive in a hover position at a safe position above the flight deck. Assisted recoveries will employ the Recovery Assist, Securing and Traversing (RAST) System in which a messenger cable from the helicopter is lowered to the ship's flight deck where it is attached to a fitting on the free end of a recovery assist cable. When the helicopter crew retrieves the messenger cable the recovery assist cable and its end fitting is drawn into and locked within the main probe of the helicopter. Below deck RAST machinery, under the control of the RAST control officer, reels in the slack recovery assist cable until taut whereupon the control system automatically maintains a constant hover tension on the cable to provide a steadying effect on relative helicopter to ship motion. The Horizon Reference Set is to be installed in the LAMPS MK III ships with the Stabilization Data Indicator mounted on, or near, the top of the hangar at the centerline and in the same plane of vision as seen by the helicopter pilot conducting his normal obstruction clearance scan during the landing approach and recovery operation. The Electronic Components Assembly is to be mounted in a protected area in the hangar or the flight control station. Where required, the Control-Indicator will be mounted in the flight control station to provide remote control of the set. Vertical reference signals will be obtained from the ships's central reference system.

6.2 Procurement requirements. Procurement documents shall specify the following:

a. Title, number and date of this specification

b. First article inspection requirements including disposition of tested equipment (see 4.2.3, 4.8 and 6.4)

c. Quality program requirements if other than in accordance with MIL-Q-9858 (see 4.1)

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d. Any deviation from production inspection requirements (see 4.2.4.1)

e. Items, or sampling basis, to be selected for production control inspections if required (see 4.2.4.2)

f. Level of preservation, packaging and packing if other than Level "A" for spares and Level "C" for complete sets (see 5.1 and 5.2.2)

6.3 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104-.9(n) of the Defense Acquisition Regulations, the data requirements identified below will be developed in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
3.3.3.4, and and 4.6.2.4	Failure Report	UDI-R-21141A
3.3.4.4	Technical Manual	UDI-M-21368
3.4	Program Parts Selection List	DI-E-7027
3.8	Request, Nomenclature and Identification Assignment and Approval	UDI-E-21336A
4.1.	Quality Assurance Program Plan	UDI-R-21374A
4.2.2 and 4.3	Test Plan (Burn-In)	DIT-5204
4.2.3 and 4.3	First Article Qualification Test Plan	DI-T-5315A
4.2.4.1 and 4.3	Acceptance Test Procedures (Production Inspection)	DI-T-3714A
4.2.3 and 4.4.2	First Article Test Report	UDI-T-23790
4.2.4.1 and 4.4.3	Acceptance Test Report (Individual Unit Production Inspection)	DI-T-3721A
4.2.4.2 and 4.4.3	Test Report (Production Control Inspection)	DI-T-2072

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

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6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulations. The first article should be one or more representative system configurations composed of sample individual items selected from the first three production systems. Individual item tests under the first article examination may be conducted on sample items selected from the first three production systems. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article.

6.5 Definitions. For purposes of this specification, definitions in MIL-STD-109, MIL-STD-280, MIL-STD-721 and the following apply.

6.5.1 Operational readiness test. That set of functional tests and observations necessary for the operating crew to determine the operational usability of the HRS.

6.5.2 Direct maintenance manhours. Direct maintenance manhours (DMMH) are the number of direct labor manhours expended on the HRS or related support equipment, including operation of the support equipment. Manhours for the following indirect activities shall not be included in DMMH: alert duty, supervision, administration, management, training, meetings, work load control, material control, travel, and any other activity inherently associated with delay time.

6.5.3 Organizational maintenance. Organizational maintenance includes all maintenance performed at, on, in or to the HRS using only those skills, tools, support equipment, publications, procedures, and techniques specified and approved for normal service use when deployed.

6.5.4 Product. Includes parts, subassemblies, assemblies, units, equipments and sets.

6.5.5 Full operating capability. The state wherein all equipment is functional and operating within specified limits.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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